

ALBION, NE, USA

WMO: 723441

Lat: 41.730N

Lon: 98.054W

Elev: 1799

StdP: 13.77

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54921

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.0	1.0	-11.3	3.2	-1.9	-6.8	4.1	3.5	33.5	25.9	30.5	26.6	6.2	330	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	93.0	73.3	90.2	72.7	87.5	72.0	77.5	87.1	75.7	85.5	74.2	83.8	11.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.6	138.7	84.2	72.7	129.5	82.3	71.4	123.7	80.7	42.4	87.4	40.5	85.9	38.9	84.2	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.0	24.0	20.4	-13.3	97.9	5.0	2.8	-16.9	100.0	-19.8	101.6	-22.6	103.2	-26.3	105.3		
			-13.8	80.5	4.9	1.8	-17.3	81.7	-20.2	82.8	-23.0	83.8	-26.6	85.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	24.8	26.8	39.6	49.3	60.0	70.7	74.9	71.7	64.6	50.7	38.7	26.7
	DBStd	20.10	12.09	12.15	11.57	9.64	8.97	6.33	5.76	5.32	7.89	9.06	10.02	11.18
	HDD50	3114	780	650	357	126	17	0	0	0	3	101	356	724
	HDD65	6351	1245	1069	787	476	208	19	5	9	103	451	790	1189
	CDD50	3114	0	1	36	105	326	621	772	674	440	123	16	0
	CDD65	876	0	0	1	5	52	190	312	218	90	8	0	0
	CDH74	8905	0	1	33	105	660	1962	3188	1872	948	129	7	0
	CDH80	3291	0	0	6	25	223	706	1347	649	299	36	0	0
	WSAvg	8.4	8.7	8.9	10.0	10.8	9.5	8.2	6.3	6.1	7.4	8.3	8.3	8.4
	PrecAvg	25.9	0.4	0.7	1.8	2.5	4.0	4.0	3.3	3.3	2.6	1.6	1.2	0.5
Precipitation	PrecMax	37.6	1.5	3.0	7.1	9.6	8.7	9.6	9.5	11.2	7.6	4.2	4.3	1.2
	PrecMin	14.5	0.0	0.0	0.0	0.5	0.4	0.9	0.4	0.4	0.4	0.0	0.0	0.0
	PrecStd	5.5	0.4	0.6	1.6	1.8	1.9	2.3	2.1	2.0	1.7	1.2	1.1	0.4
	PrecStd	5.5	0.4	0.6	1.6	1.8	1.9	2.3	2.1	2.0	1.7	1.2	1.1	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.6	65.9	78.8	83.5	91.5	95.0	97.7	94.7	91.2	85.9	73.4	58.8
		MCWB	45.7	48.4	55.7	61.7	66.2	72.4	73.7	74.1	70.7	65.5	53.7	45.0
	2%	DB	50.9	56.4	71.2	76.7	86.1	90.9	94.1	90.1	87.6	77.0	66.4	52.0
		MCWB	40.8	44.9	53.6	57.5	66.2	71.3	74.1	74.5	70.8	59.7	51.3	42.6
	5%	DB	45.1	49.9	64.4	71.6	81.3	87.7	90.7	86.4	83.5	72.4	60.7	46.0
		MCWB	37.8	41.1	51.8	56.0	63.1	70.4	74.5	73.7	68.9	57.3	48.2	39.0
	10%	DB	40.7	43.3	58.7	66.5	76.8	83.9	87.7	83.6	79.9	67.7	54.6	41.2
		MCWB	35.6	37.0	48.1	53.4	61.7	69.0	73.9	71.9	66.2	55.3	45.3	36.0
	0.4%	WB	45.6	50.2	61.5	63.8	71.1	78.3	80.4	79.5	75.6	68.2	57.2	48.0
		MCDB	58.4	62.6	69.9	75.5	84.0	88.6	90.5	88.6	85.2	78.1	67.8	55.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.4	45.1	57.2	60.3	68.5	75.1	78.1	76.9	73.0	63.4	52.9	42.8
		MCDB	49.9	56.1	66.9	72.3	81.2	85.9	88.0	86.1	82.8	72.8	64.5	51.1
	5%	WB	38.3	41.2	52.6	57.4	66.0	73.0	76.4	75.1	71.0	59.8	49.5	39.5
		MCDB	44.1	48.6	63.1	68.0	77.3	83.7	86.3	83.8	79.7	67.5	59.8	45.5
	10%	WB	35.7	37.5	48.2	54.9	63.4	71.0	74.7	73.3	68.8	56.6	45.8	36.4
		MCDB	40.4	42.9	58.8	64.8	74.2	80.5	84.8	81.4	77.0	65.5	54.4	41.1
Mean Daily Temperature Range	5% DB	MDBR	19.5	20.0	22.5	22.8	22.4	21.8	21.5	20.6	24.2	24.3	23.5	19.0
		MCDBR	26.3	29.1	33.6	31.3	29.2	25.6	25.3	24.3	27.1	32.7	33.2	26.3
		MCWBR	17.7	18.1	18.3	16.4	13.0	10.7	11.0	12.1	12.7	16.9	19.1	18.1
	5% WB	MCDBR	24.3	27.9	29.3	26.7	24.6	22.2	21.6	21.3	22.4	25.0	30.6	24.6
		MCWBR	17.2	18.1	17.3	15.2	12.1	11.4	11.2	11.7	12.1	14.9	18.5	17.4
Clear-Sky Solar Irradiance	taub		0.274	0.282	0.307	0.355	0.376	0.395	0.400	0.404	0.363	0.324	0.281	0.270
	taud		2.462	2.475	2.443	2.320	2.346	2.339	2.345	2.321	2.393	2.484	2.516	2.478
	Ebn at Noon		282	297	300	290	285	279	276	272	276	277	277	274
	Edn at Noon		23	27	32	39	39	39	39	39	33	27	22	21
	RadAvg		664	952	1255	1599	1866	2128	2124	1815	1473	1040	732	554
All-Sky Solar Radiation	RadStd		57	56	107	90	129	144	98	123	96	111	54	53

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page